TITLE PAGE

- **Problem Statement ID -** SIH26034
- **Problem Statement Title -**Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels
- **Theme -** Miscellaneous
- **PS Category -** Software
- **Team ID -** 158672
- **Team Name –** TEAM TG PRP

TEAM
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AI-Powered Legal Metrology & Packaged Commodity Compliance System

Transforming Legal Metrology Compliance through Multi model AI Inspection

SMART INDIA HACKATHON 2026

❖ Proposed Solution :-

- The proposed solution uses advanced computer vision and machine learning models to inspect packaged commodities based on visual data, influencing factors such as **text, regulatory logos, barcodes and cross modal tamper detection.**

- A user friendly website allows officers to easily capture product images and view compliance result, it features an **offline first approach,** ensuring seamless data storage in remote areas and automatic synchronization when internet connectivity returns

- The solution combines data driven AI model for accurate legal compliance checks and has a user friendly and scalable web interface for efficient field to desk operations.

❖ Uniqueness :-

1) Computer Vision Based text extraction and Measurement :-

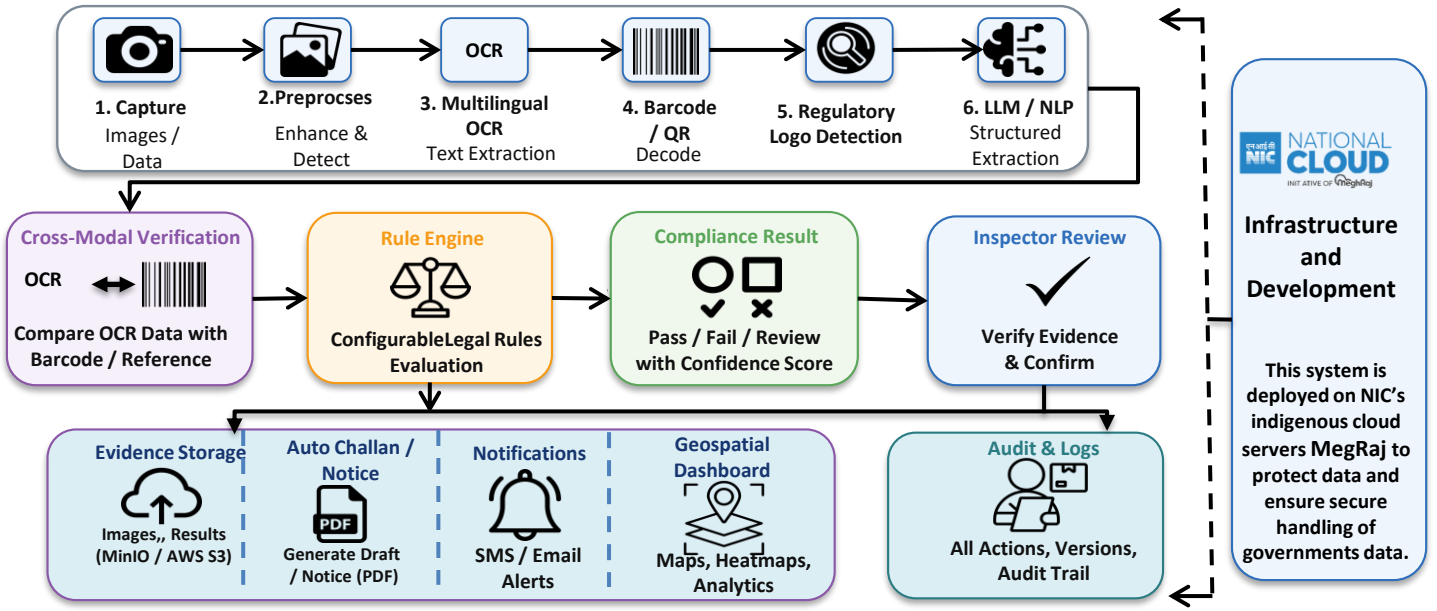
Using ORC and ArUco reference tag to extract and measure font size in real word millimeters not just pixels

2) Deterministic Legal Rule Engine :-

Use Programed blocked rule engine for fixed explainable legal decisions no LLM hallucinations in penalties or compliance result

3) Offline First and Tamper Proof Inspection :-

Captures data off-line without internet, lock images using SHA-256 hashing and automatically syncs data when connectivity returns



TECHNICAL APPROACH

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Rule 19 & Rule 21 : Premises Context

Inspections must be tied to a physical location (Factory under Rule 19, or Retail Shop under Rule 21)

Location Data
Latitude, Longitude

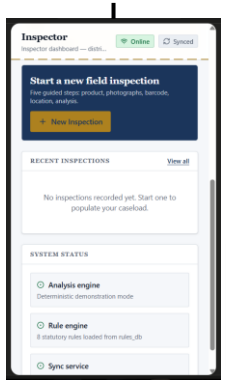
Rule 11, 12 & 13 : Standard Units & Phrasing

Quantity must be in metric units (g, kg, ml, L). Vague terms like "approx" or "minimum" are strictly banned.

Rule 6(1)(a to f) : Mandatory Declarations

Label must show Manufacturer Name/Address, Generic Name, Net Qty, Date, MRP (incl. of all taxes), and Consumer Care details.

2.FRONTEND



React/Next.js

Website behaves like mobile app — PWA

Inspection map — leaflet

Google Map Apl

INSPECTOR LOGIN



1. Related Details
2. Capture Images
3. Submit

Products Inspection

Overall Data

interconnected

ADMINISTRATION LOGIN

1. HEAT Map
2. Approval System
3. Show KPI'S

Inspector After Submition

5.OUTPUT
INSPECTOR Verificitation Result

1. Verification
2. Report Generate
3. Rule Evaluation
4. Violations Detected
5. Evidence
6. Inspector & Administrator Approval

Pass:certificate Generate

Fail:Legal notice generated along with Chalan

PostgreSQL
Store structured information

PostGIS
Store Inspection Location Coordinates

Hfont,mm=Hchar,px/PMM

$H_{font,mm} \geq H_{statutory,min}$ = **PASS**

$H_{font,mm} < H_{statutory,min}$ =**FAIL**

4.EXTERNAL API SERVICES

- **fssai** — License Verification API
- **GST** — GST Verification API
- **Google Maps** — Address / Location API
- **GS1** — Product Reference Data API

1. DATA CAPTURE LAYER

- **Capture** — 1.Front Image 2.Back Image 3.Close-up
- **Barcode /QR Scanner**
- **GPS + Location Shop Metadata**
- **BHASHINI** — Regional Language UI
- **Metadata** — extra details
- **offline Data** — Store locally

Rule 7 (Table-I & II) N : Numeral & Font Height

PMM=Wmarker_px/Wmarker_mm

Physical Font Height Calculation



Rule 22:package's net weight falls

Allowed Deficit
=Declared Quantity × (MPE%/100)

Minimum Legal Weight
=Declared Quantity−Allowed Deficit

$C_W = \frac{(I_{text} - I_{bg})}{I_{bg}}$

Ensures text stands out clearly against the package background



Rule 18(2) & 18(5) : Retail Overpricing & Tampering

A dealer cannot alter, smudge, or paste a new sticker over the printed MRP, nor sell above it.

MongoDB
Data Store JSON Format

Text must "contrast conspicuously" with the background. The area around the Net Qty must be clear of graphics.

Rule 8 & 9 : Legibility & Contrast

3.BACKEND

API GATEWAY

- Node.js/Express.js
- Redis
- Websocket
- status
- live
- frontend

- Capture Image — OpenCV
- Detect marks/symbols — YOLOv8
- Open Source LLM — Row text convert json
- PaddleOCR — OCR Text Extraction
- Pyzbar — QR code read/detect

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FEASIBILITY AND VIABILITY

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FEASIBLITY

Technical Feasibility

Built using modern open source technologies (Next.js,FastAPI,YOLOv8, PaddleOCTR,supabase) offline first PWA with seamless sync

Operational Feasibility

Simple workflow for inspectors with mobile app and dashboard, Automated AI and Rule Based validation reduces manual efforts and errors

Economic Feasibility

Open-source AI tools and cloud ready infrastructure reduce development Offloading heavy image streams to Redis task queues prevents the server from blocking and reduces cloud hosting costs.

Regulatory Feasibility

Support **Legal Metrology Rules, 2011 Act ,FSSAI ,GST** and packaging rules

Scalability & Future Readiness

Modular design allows addition of new APIs and models. Can scale from SIH MVP to enterprise use

CHALLENGES

AI Hallucination & False Positives

TECHNICAL MITIGATION

Deterministic Rule Engine BRMS

Separation of AI extraction from hard coded Python legal rules ensures 100% compliant, predictable outcomes.

Poor Field Connectivity

TECHNICAL MITIGATION

Offline-First PWA Queue

Utilizes Indexed DB storage with secure client UUIDs and robust auto-sync upon connection restoration.

Extract Physical Measurements from image

TECHNICAL MITIGATION

Using reference ArUco Marker

Optical Metrology (PMM Calibration): OpenCV detects a 20mm ArUco Reference Marker. Calculates Pixels per mm (PMM). Divides text pixel height by PMM to check physical compliance in millimeters

PMM=Wmarker_mm/Wmarker_px

High Server Load / Traffic Spikes

TECHNICAL MITIGATION

Redis In Memory Buffering

Leverages Fast API HTTP 202 async task dispatch to securely decouple request intake from intensive processing.

Integrating into Existing Infrastructure &Ensuring Adoption

TECHNICAL MITIGATION

1. Language Barriers

Multi-lingual support (Hindi + Regional languages via Bhashini), • Local UI for inspectors

2. System Compatibility

Works with existing devices and network conditions Cross platform support (Android/iOS/Web)

OVERALL VIABILITY

Technically sound | Operationally practical | Economically efficient | Legally aligned | Ready to scale

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IMPACT AND BENEFITS

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IMPACT ON TARGET AUDIENCE

Our solution empowers all key stakeholders in the packaged commodity ecosystem.

Inspectors & Field Officers
- Faster & accurate inspections
- Offline access in remote areas
- Easy evidence & report generation

Consumers
- Safer and genuine products
- Protection from fraud
- Greater trust and transparency

Businesses & Manufacturers
- Fair competition
- Reduced counterfeit risk
- Better brand reputation

Government & Regulators
- Efficient compliance monitoring
- Data-driven decision making
- Stronger enforcement

BENEFITS OF THE SOLUTION

SOCIAL BENEFITS
Safer Communities, Healthier Lives

- Reduces counterfeit and substandard products in the market
- Protects consumer health and safety
- Builds trust in packaged commodities

Healthy people. Stronger communities.

ECONOMIC BENEFITS
Fair Market, Stronger Economy

- Prevents revenue loss due to tax evasion and under-declaration
- Supports honest manufacturers and small businesses
- Reduces cost of compliance through digital automation

More compliance. More growth.

ENVIRONMENTAL BENEFITS
Sustainable Choices, Greener Future

- Promotes responsible packaging and labeling
- Reduces waste from fake/expired products
- Supports sustainable and eco-friendly supply chains

Less waste. A cleaner tomorrow.

OPERATIONAL BENEFITS
Smarter Systems, Higher Efficiency

- Automates manual and repetitive tasks
- Saves time with AI-powered analysis
- Enables offline-first field operations

Do more. In less time.

THE BIG PICTURE

Empowering inspectors, protecting consumers, strengthening businesses and building a safer, fairer and more sustainable India.

Better Compliance Today for a Stronger India Tomorrow

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RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

LM Legal Metrology & Compliance (India)

- Legal Metrology (Packaged Commodities) Rules, 2011 Click
- 2026 Amendment Rules – AEO & Corporate Accountability Click

CV Computer Vision for Regulatory Marks & Food Safety

- 5-Class Food Certification Dataset Click
- AI in Food Safety Governance – Review Article Click
- YOLO: Unified, Real-Time Object Detection – Original Paper (arXiv) Click
- Ultralytics YOLOv8 – Official Documentation Click

AI Structured Extraction (LLM/NLP) & Multilingual OCR for Indian Languages

- Llama 2: Open Foundation and Fine-Tuned Chat Models (arXiv) Click
- pyzbar – Barcode/QR Decoding Library Docs (PyPI) Click
- PaddleOCR-VL: Multilingual OCR with 109 Languages Click
- PaddleOCR PP-OCRv5 Multilingual Models Click

Live Demo https://vyapar-nirikshan-portal.onrender.com

GitHub Repo https://github.com/Tushar-Ganmukhe/SIH-26.git

Demo Video https://youtu.be/cCVw9smMn3c

❖ **COMPARISON TABLE WITH EXISTING SYSTEMS :-**

Sr	Feature	Our System	Existing System (wooqer)
1	Legal Metrology Enforcement	✓	✗
2	Automated Label Measurement	✓	✗
3	Deterministic Rule Engine	✓	✗
4	Government Verification	✓	✗
5	Workflow / Corrective Action	✓	✓
6	Inspection Records	✓	✓

DATA Food Packaging & Label Datasets

- General Food Packaging Recognition Click
- Package & Label Detection Click
- Food Packaging Info (Text & Regions) Click

Measuring Measuring Size of an Object using Computer Vision

- International Journal of Innovative Technology and Exploring Engineering (IJITEE) ISSN: 2278-3075, Volume-8, Issue-6S4, April 2019